Comparison of Gaussian Process Covariance Kernels for Bayesian Optimization of High-Order Harmonic Generation

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Here, we compare the performance of four covariance kernels for 2D Bayesian optimization using the LogEI acquisition function in the transformed optimization space $(\log(pL), \log(p/L))$, namely Matérn 1/2, Matérn 3/2, Matérn 5/2, and the radial basis function (RBF). These kernels encode differing assumptions regarding the smoothness of the Gaussian Process (GP) surrogate. The general Matérn covariance function is given by

$$K_{\theta,\nu}(\mathbf{x}, \mathbf{x}') = s^2 \frac{2^{1-\nu}}{\Gamma(\nu)} \left(\sqrt{2\nu}\, R_{\boldsymbol{\ell}}\right)^{\nu} K_{\nu}\left(\sqrt{2\nu}\, R_{\boldsymbol{\ell}}\right), \tag{1}$$

where $R_{\boldsymbol{\ell}}$ is the scaled Euclidean distance between two points $\mathbf{x} = (x_1, \ldots, x_D)$ and $\mathbf{x}' = (x'_1, \ldots, x'_D)$ in the D-dimensional parameter space, given by

$$R_{\boldsymbol{\ell}} = \sqrt{\sum_{j=1}^{D} \frac{(x_j - x'_j)^2}{\ell_j^2}}, \tag{2}$$

where ℓ_j is the characteristic length scale associated with coordinate x_j, s^2 is the output variance, Γ is the gamma function, and K_ν denotes the modified Bessel function of the second kind of order ν. The parameter ν determines the smoothness of the GP prior. The Matérn 1/2, Matérn 3/2, and Matérn 5/2 kernels correspond to $\nu = 1/2$, 3/2, and 5/2, respectively, whereas the RBF kernel is recovered in the limit $\nu \to \infty$, corresponding to an infinitely differentiable covariance function.

As shown in Figure 1a, the Matérn 3/2, Matérn 5/2, and RBF kernels exhibit nearly identical performance in terms of optimization convergence. Although the Matérn 3/2 kernel shows faster initial convergence, all three reach the same best-so-far observed conversion efficiency $\mathrm{CE}_{\mathrm{best}}$ by iteration 31. The Matérn 1/2 run was discontinued after iteration 10 because GP fitting failed, preventing a comparison of its long-term performance.

In terms of normalized root-mean-square error (NRMSE) and structural similarity index measure (SSIM) evaluated against the reference 100×100 grid scan (Figure 1b and c), the three completed runs achieve good surrogate reconstruction, although the RBF kernel exhibits somewhat lower predictive accuracy than the Matérn 3/2 and 5/2 kernels. The latter two perform particularly similarly, with the Matérn 5/2 yielding slightly better final accuracy: NRMSE of approximately 0.039 versus 0.045, and SSIM of approximately 0.985 versus 0.976, respectively. Consequently, the Matérn 5/2 kernel was retained for the main optimization

studies as a robust choice, combining strong optimization convergence with the most accurate final surrogate reconstruction. While the Matérn 3/2 kernel converges somewhat faster initially, the differences between the two are small, and the RBF kernel remains a competitive alternative.

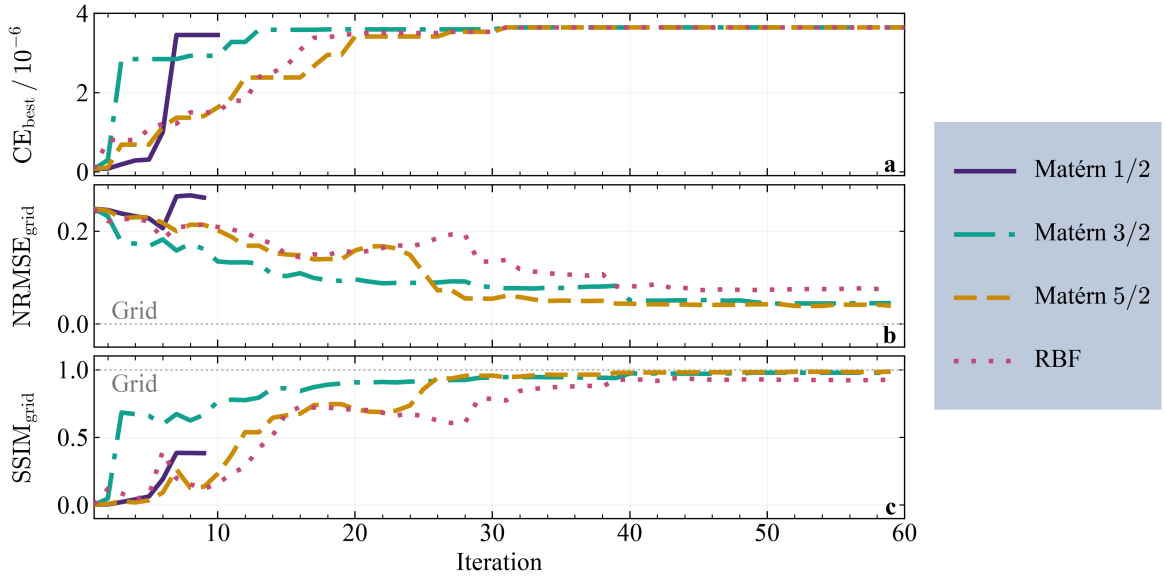


Figure 1: **Convergence and GP prediction accuracy for different kernels. a)** Evolution of the best-so-far CE as a function of the BO iteration. **b)** NRMSE between the GP prediction and the direct grid scan ($NRMSE_{grid}$) as a function of the BO iteration. The dotted gray horizontal line indicates the value of 0, corresponding to perfect agreement with the direct grid scan. **c)** SSIM between the GP prediction grid and the direct grid scan ($SSIM_{grid}$) as a function of the BO iteration. The dotted gray horizontal line indicates the value of 1, corresponding to perfect agreement with the direct grid scan. Different colors and marker styles distinguish the kernels.